A Visual Trace Map of Edinburgh Place Names in Literature

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Abstract

Research Ethics Approval

This project obtained approval from the Informatics Research Ethics committee.
Ethics application number: 446635
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Declaration

I declare that this thesis was composed by myself, that the work contained herein is my own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

(*Merritt Wang*)

Acknowledgements

Table of Contents

List of Figures

Chapter 1

Introduction

1.1 Motivation

Edinburgh is among the most literarily documented cities in the world. The Palimpsest project at the University of Edinburgh systematically extracted place-name mentions from hundreds of literary works spanning three centuries, producing the LitLong Edinburgh database: 620 documents published between 1687 and 2015, 2,135 distinct place names, and 50,248 geo-referenced mention records [1]. The project made this corpus publicly accessible through a web interface that plots place mentions on an interactive geographic map.

Geographic maps are a natural starting point for literary data of this kind, but they carry an implicit assumption: that the meaning of a place name is determined by its location. In geography, this is uncontroversial. In literature, it is not. When Irvine Welsh names Leith in *Trainspotting*, the word does not primarily refer to a set of coordinates north of Edinburgh city centre. It evokes a social world, a class position, a political history. When Walter Scott names the Canongate, he is invoking a specific stratum of Edinburgh's past. A geographic map can show *where* these places are. It cannot show *when* in the narrative they appear, *how often* they recur, *which other places* surround them, or *whose authorial voice* has chosen them.

This project proceeds from the observation that place names in literary text are data about narrative structure as much as about geographic space. The visualisation problem, accordingly, is not how to render literary geography more accurately on a map, but how to represent narrative structure in a form that reveals patterns invisible to geographic tools.

1.2 Problem Statement

The core question this dissertation addresses is: can place names in literary text be visualised as an independent narrative structure, rather than as a rendering of geographic information?

This question divides into two sub-problems corresponding to two distinct dimensions of narrative structure. The first concerns *authorial spatial attention*: does each author's distribution of place-name mentions across Edinburgh constitute a distinctive pattern that can be recognised without prior knowledge of the author? The second concerns *narrative co-occurrence*: do places that appear together across literary works form a topology that differs systematically from their geographic arrangement?

Existing tools do not address either question. They do not separate literary space from geographic space. They do not support comparison of authorial spatial styles. And they do not reveal which places are narratively proximate, as opposed to geographically proximate.

1.3 Research Questions

Two formal research questions guide the project:

RQ1 (Author Spatial Fingerprint): Does each author's distribution of place-name mentions across Edinburgh's official neighbourhood sectors form a visually distinctive spatial signature, sufficient for users to identify authors from their fingerprint alone?

RQ2 (Narrative Topology): Does the co-occurrence of place names within literary works reveal a narrative topology that diverges meaningfully from the geographic topology of Edinburgh?

The success criterion for RQ1, established in discussion with the supervisor, is that different authors must produce visually distinct fingerprint shapes on a one-to-one basis: no two authors should produce shapes that are indistinguishable to a naive user. The success criterion for RQ2 is that the visualisation should surface cases where narrative proximity contradicts geographic proximity—places that are spatially distant yet repeatedly appear together in the same literary works.

1.4 Contributions

This dissertation makes the following contributions:

1. A novel interactive visualisation system for literary place-name data that represents narrative structure rather than geographic location, implemented in D3.js and delivered as self-contained, deployable HTML files.

2. Two complementary visualisation directions: author spatial fingerprints (three designs) and narrative topology networks (three designs), each driven by real data from the LitLong Edinburgh corpus.

3. A validated spatial classification of Edinburgh place names using official administrative boundary data from the City of Edinburgh Council, comprising 14 sectors derived from the Neighbourhood Partnership Areas and Natural Neighbourhoods datasets (Open Government Licence v3.0).

4. An empirical finding with implications for literary geography: among five representative Edinburgh authors, no single place name is mentioned by three or more authors, and only 18 are shared by any two. Each author constructs an almost entirely private imaginary Edinburgh.

5. A method for extracting narrative reading order from the LitLong database by treating the `start_word` field as a globally monotonic index, enabling position-normalised analysis of when and where in a narrative each place appears.

6. A user evaluation comparing domain expert and general public responses to the six visualisation designs, informing the final design selection.

1.5 Dissertation Overview

Chapter 2 situates the project within three bodies of literature: literary geography and geocriticism, existing visualisation approaches for literary corpora, and information visualisation principles for humanities data. Chapter 3 describes the data exploration process, the design decisions made for each of the six visualisation candidates, and the rationale for the choices made. Chapter 4 covers the technical implementation, from database setup through to the D3.js interactive prototypes. Chapter 5 describes the user study design and reports its results. Chapter 6 discusses the findings in relation to the research questions and reflects on the project's limitations. Chapter 7 concludes and outlines directions for future work.

Chapter 2

Literature Review

2.1 Literary Geography and Geocriticism

The relationship between literature and place has been a concern of literary scholarship since at least the late twentieth century, but the theoretical grounding for a spatially-oriented reading of fiction consolidated around what is now called the "spatial turn" in the humanities [9]. Bertrand Westphal's geocriticism argues that literary places are not simple reflections of real geographies but active constructions: representations that reshape the reader's understanding of space and that exist in dialogue with the physical world rather than merely depicting it [11]. On this account, the place names in a novel are not pointers to locations on a map but nodes in a narrative network.

Franco Moretti's quantitative approach to literary geography offered a different perspective [5]. By mapping the settings of hundreds of European novels, Moretti demonstrated that literature does not distribute itself uniformly across geographic space: certain places are obsessively returned to, others systematically avoided. He coined the phrase "literary silences" to describe the latter—absences that are as informative as presences. Moretti's maps are geographic, but his interpretive framing is already narrative: he reads the maps for patterns of attention and neglect, not for spatial accuracy.

Both traditions converge on a point that motivates this project: place in fiction is a narrative construction, not a geographic fact. The visualisation problem is therefore not how to represent the geography of literary Edinburgh more faithfully, but how to make the narrative dimensions of that geography—frequency, authorial style, co-occurrence, temporal position—visible.

2.2 The LitLong Edinburgh Project

The empirical foundation of this project is the LitLong Edinburgh corpus, produced by the Palimpsest project at the University of Edinburgh [1]. Using natural language processing pipelines, the project extracted geo-referenced place-name mentions from a curated selection of literary works set in or concerning Edinburgh, spanning the period 1687 to 2015. The full database—accessed here via a PostgreSQL dump released in 2022—contains 620 documents, 2,135 distinct place names, 50,248 mention records, and 424 authors, across 29 literary genre classifications.

Several features of the database are particularly relevant to this project. The `api_sentence` table stores the complete source sentence for every mention, enabling retrieval of the exact literary context in which a place name appears. The `start_word` field in `api_locationmention` encodes a global word index that is strictly monotonically increasing within each document, providing a proxy for narrative position without requiring full text parsing. The `api_posmention` table contains over two million part-of-speech annotations, potentially supporting future analysis of whether a place name appears in dialogue, narration, or description—a dimension not explored in this dissertation.

The original LitLong web interface renders place mentions as dots on an interactive geographic map. This design, while intuitive and technically accomplished, makes the assumptions critiqued above: it positions geographic location as the primary axis of literary meaning. The present project treats the LitLong data as the raw material for an alternative representational strategy.

2.3 Visualising Place in Literary Corpora

The closest antecedent to the present project is the work of Stange and Dörk on visualising literary place mentions in novels set in Berlin and New York [8]. Their "Novel City Maps" approach generated fingerprint-like representations of each novel's spatial distribution across the city, using street-map coordinates as the positional axis. The present work extends this approach in a significant direction: rather than retaining geographic coordinates, it replaces them entirely with an abstract categorical axis (city sectors), allowing the fingerprint shape to represent authorial spatial attention without the perceptual noise introduced by real-world topography. This shift is motivated by the data: as Section 3.2 reports, place names in the LitLong corpus are too sparsely shared across authors for individual place names to serve as meaningful axes.

Other digital literary geography projects have largely retained the geographic map as their primary interface. Literary Atlas, Google's Literary Trips, and various GIS-based academic projects all plot textual mentions against spatial coordinates. While these tools are valuable for exploring the geographic distribution of literary references, they cannot address questions about authorial style, narrative co-occurrence, or the divergence between spatial proximity and narrative proximity.

The information visualisation literature offers relevant precedents for the two directions pursued here. Shneiderman's small multiples principle [10]—presenting multiple instances of the same visual form side by side for comparison—underlies the Small Multiples design in Direction 1. The force-directed graph layout used in Direction 2 follows Fruchterman and Reingold's energy-minimisation algorithm [4]. The metro-style map adapts Beck's insight that functional topology—the traveller's need to know which line and which direction—can be more usefully represented than geographic accuracy [2].

2.4 Information Visualisation for Humanities Data

Shneiderman's information-seeking mantra—"overview first, zoom and filter, details on demand" [7]—provides a structural principle for the interaction design in this project. All six visualisation prototypes follow this pattern: an overview of the full dataset (all authors, all co-occurrence pairs) is presented by default, with interactive mechanisms for filtering and zooming, and detail panels that surface place-specific and book-specific information on demand.

Drucker argues that humanistic data should be understood as *capta*—actively interpreted and constructed—rather than as objectively given [3]. This perspective is relevant to several design decisions documented in Chapter 3: the choice of sectors as axes reflects an interpretive act (what counts as a meaningful spatial unit for literary analysis?), and the choice of document-level co-occurrence as the granularity for the topology network reflects a judgment about what constitutes meaningful narrative co-presence.

Tufte's principle of small multiples [10] provides the theoretical foundation for the side-by-side comparison panel in Direction 1, where five maps of the same spatial extent but different author-specific bubble distributions enable direct visual comparison of authorial spatial patterns.

2.5 Graph and Network Visualisation

The force-directed network visualisation in Direction 2 follows the Fruchterman-Reingold algorithm [4], which models graph layout as a physical simulation in which nodes repel each other and edges act as springs. The critical property of force-directed layouts for this project is that node positions are determined entirely by the topology of the co-occurrence network, not by geographic coordinates. A node's position in the final layout reflects its narrative connectivity, not its map location.

Nöllenburg and Wolff's work on algorithmic metro-map layout [6] establishes that schematic map drawing—placing nodes on an octilinear grid while preserving topological correctness and legibility—is a computationally hard problem in the general case. An initial prototype of the metro-style visualisation sidestepped this problem by hand-drawing eight lines grouped by geography; a later audit (Section 3.5.0.0.1) showed that this hand-drawn grouping was only weakly supported by the measured co-occurrence data. The metro-style visualisation was therefore rebuilt around a heuristic pipeline—community detection on the co-occurrence graph, a co-occurrence-weighted spring embedding, and a local angle-correction pass to approximate octilinearity—that does not solve Nöllenburg and Wolff's problem exactly but derives the line structure and station layout from data rather than manual judgment.

2.6 Gap and Contribution

The literature reviewed above identifies three gaps that this project addresses. First, while literary geography theory has established that place in fiction is a narrative construction, existing digital tools continue to render literary place names as geographic data. Second, no existing tool supports direct visual comparison of authorial spatial styles as abstract fingerprints. Third, no existing tool visualises the co-occurrence topology of literary place names as a structure distinct from geographic proximity. This project addresses all three gaps through a system of interactive visualisations grounded in real corpus data and evaluated through a user study.

Chapter 3

Design

3.1 Design Rationale

The project brief specified D3.js as the implementation technology, and this choice aligns with the requirements of the visualisation task. D3.js is web-native, requiring no installation or server infrastructure for end users; this matters for sharing prototypes with supervisors and study participants. It provides fine-grained control over visual encoding, enabling the custom interactive designs required by both research directions. It is the standard tool in digital humanities visualisation practice, ensuring that the implementation approach is reproducible by others in the field.

The choice between static and interactive visualisation was settled early in the project. The LitLong corpus contains 620 documents, 2,135 place names, and 424 authors. A static visualisation that attempted to show all of this simultaneously would be unreadable. Interactivity is necessary to support the "overview first, zoom and filter, details on demand" pattern [7] that allows a reader to form a general impression and then drill into specific authors, sectors, or co-occurrence pairs.

The overarching design philosophy is that narrative structure should take precedence over geographic accuracy. Where geographic coordinates are used (the Small Multiples view), they serve as a spatial anchor to help users orient themselves, not as a claim that geographic position determines literary meaning. All other designs deliberately abandon coordinates in favour of abstract encodings that foreground narrative properties: sector-based fingerprints for Direction 1, and co-occurrence networks for Direction 2.

3.2 Data Exploration and Key Findings

Before designing the visualisations, a series of exploratory analyses of the LitLong data were conducted to establish what the data could support. These analyses produced five findings that directly shaped the design decisions documented below.

Finding 1: Author spatial languages are highly individual. Among the five test authors selected for prototyping (Alexander McCall Smith, Irvine Welsh, John Gibson Lockhart, Walter Scott, and Robert Louis Stevenson), no single place name is mentioned by three or more authors. Only 18 place names are shared by any two. This near-total absence of overlap confirms the literary-geographic intuition that each author constructs a largely private imaginary Edinburgh, and it has a direct implication for visualisation design: individual place names cannot serve as comparable axes across authors, because most axes would be zero for most authors. This finding motivated the decision to use city sectors as the radar chart axes.

Finding 2: Sector distributions cross-validate against literary knowledge. The sector-based analysis yields distributions that align with established literary-geographic knowledge for all five test authors: McCall Smith concentrates in New Town (43.6%), consistent with his Scotland Street series; Welsh concentrates in Leith (44.3%), consistent with *Trainspotting*; Scott concentrates in Old Town (32.0%), consistent with his historical fiction; and Stevenson shows a notable Pentlands component (17.0%), consistent with his documented connection to Swanston. This cross-validation strengthens confidence in the sector classification methodology.

Finding 3: Co-occurrence granularity requires document-level analysis. Sentence-level co-occurrence analysis yields zero pairs, because no sentence in the LitLong corpus contains more than one location mention. Page-level co-occurrence yields up to 92 co-occurring places on a single page, producing graphs too dense to be readable. Document-level co-occurrence—counting pairs of places that appear in the same work—yields 403 pairs with weight ≥ 2, where weight counts the number of distinct works in which both places appear. This granularity was selected as the appropriate level of analysis for the topology network.

Finding 4: Narrative topology diverges from geographic topology. The strongest co-occurring place pair is Leith and Princes Street, with a co-occurrence weight of 28: they appear together in 28 distinct literary works. Geographically, Leith and Princes Street are separated by approximately 2.5 kilometres; Leith is a port district to the north, Princes Street the main commercial thoroughfare of the city centre. Their narrative proximity is not a reflection of their spatial proximity. This finding provides the clearest empirical illustration of the central claim of Research Question 2.

Finding 5: Walter Scott's broad geographic range creates a display problem. Scott's works reference locations far outside the Edinburgh city centre, including Forth (latitude 56.05), Haddington (longitude -2.80), and Linlithgow (longitude -3.58). When these points are plotted against a common coordinate scale in the Small Multiples view, they compress the city-centre data for all five authors into an unreadable band. The decision was made to fix the display range to the Edinburgh core area (latitude 55.85–56.02, longitude -3.35 to -3.05) and exclude out-of-range points from the map panel, while retaining them in the sector-based analyses.

3.3 Spatial Classification: Neighbourhood Sectors

A key methodological decision in Direction 1 was how to define the spatial units (sectors) that serve as axes in the radar chart and as ordering categories in the bar-code view. Three requirements guided this decision: the sectors must be few enough to serve as legible visual axes; they must be grounded in an authoritative external source so that the classification can be justified independently of the visualisation; and they must carry cultural and historical meaning relevant to literary analysis.

The solution adopted is a two-level framework derived entirely from official City of Edinburgh Council data, accessed via the Council's Open Spatial Data Portal (Open Government Licence v3.0). The primary layer is the **Neighbourhood Partnership Areas** (NP Areas), a set of 12 administratively defined zones used by the Council and its statutory partners for service planning and community consultation. These boundaries are legally established and publicly documented.

One NP Area—City Centre—was subdivided using a second official dataset, the **Natural Neighbourhoods** layer. Natural Neighbourhoods are resident-defined neighbourhood boundaries, created in 2004 as part of a ward boundary review and updated through a public online consultation in 2014. The City Centre NP Area encompasses two zones that are historically and culturally distinct in the Edinburgh literary tradition: Old Town (the medieval city, characterised by the High Street, the Canongate, and the closes of the Royal Mile) and New Town (the Georgian planned city built to the north from the 1760s onwards). These two areas are jointly inscribed as a UNESCO World Heritage Site, and their literary characters differ substantially: Old Town is the setting for the historical fiction of Scott and the Gothic sensibility of Stevenson and Hogg; New

Town is the milieu of McCall Smith's contemporary domestic fiction and Lockhart's nineteenth-century biography. Conflating them into a single City Centre sector would erase the most significant literary-geographic distinction in the corpus. A third Natural Neighbourhood unit, Canongate, was also retained as a separate sector, as it forms the eastern end of the Royal Mile and carries distinct historical significance.

The resulting framework comprises **14 sectors**: 11 unchanged NP Areas (Almond, Craigentinny/Duddingston, Forth, Inverleith, Leith, Liberton/Gilmerton, Pentlands, Portobello/Craigmillar, South Central, South West, and Western Edinburgh) plus Old Town, New Town, and Canongate derived from Natural Neighbourhoods. Place-name assignment proceeds by point-in-polygon intersection using the Shapely library. Of the 2,135 place names, 1,480 (69%) fall precisely within a sector polygon; 423 (20%) lie in inter-polygon gaps within the Edinburgh bounding box and are assigned to the nearest sector; and 232 (11%) lie outside Edinburgh entirely and are excluded from sector-based analyses. The resulting classification is cited as: "Copyright City of Edinburgh Council, contains Ordnance Survey data © Crown copyright."

3.4 Direction 1: Author Spatial Fingerprint

Three visualisation designs were developed and implemented for Direction 1.

Radar chart (primary design). The radar chart represents each author as a polygon across the 14 sector axes, where the radius at each axis encodes the proportion of that author's total mentions that fall within the corresponding sector. The polygon shape is the author's spatial fingerprint. The chart was scaled to the maximum value in the dataset (0.436, McCall Smith's New Town proportion) rather than to 1.0, because fixing the axis to 100% leaves over 60% of the radial space empty and compresses the meaningful variation into a small central region.

The primary interaction is point-level: hovering over any vertex reveals the sector percentage and a ranked list of the places and books that contribute to it. This responds directly to the supervisor's feedback that the visualisation should provide "low-level information upon interaction." A detail panel opens on click, listing the top place names and top books in the selected sector for the selected author; hovering over an individual place row within the panel additionally surfaces a real example sentence drawn from `api_sentence`, together with the book and author it is drawn from. The legend supports filtering by author (click to hide/show) and isolation (double-click to show only one author), enabling the user to compare any pair or subset of authors.

Bar-code fingerprint (secondary design). The bar-code view represents each author as a horizontal row of vertical bars, one bar per place name, ordered from left to right by sector (north to south). The height of each bar encodes the proportion of that author's total mentions attributable to the corresponding place.

A key design decision concerned the Y-axis scaling. Welsh's Leith accounts for 44.3% of his total mentions, substantially more than any single place for any other author. On a shared linear Y-axis scaled to this maximum, all other authors' distributions are compressed into a band occupying less than a quarter of the chart height, rendering within-author variation imperceptible. The solution adopted is a per-author independent Y-axis, where each row is scaled to its own maximum. This design choice prioritises the visibility of within-author spatial structure—the *shape* of each author's distribution—over cross-author magnitude comparison. The trade-off is that bar heights cannot be directly compared across rows; this is made explicit by a warning note in the interface. A shared-axis mode (scaling to the global maximum) remains available via a toggle, allowing users to make cross-author magnitude comparisons when needed. The bar-code view underwent three iterations of scale design before reaching its final form. The first prototype used a logarithmic scale, following standard practice for long-tail frequency distributions. However, supervisor feedback noted that logarithmic scales are difficult for non-specialist users to interpret intuitively, and that the shape of the distribution—not its absolute magnitude—was the primary object of interest. The second prototype adopted a linear scale shared across all five authors, but this introduced a new problem: Welsh's Leith accounts for 44.3% of his total mentions, substantially more than any single place for any other author. On a shared linear Y-axis, all other authors' distributions were compressed into a band occupying less than a quarter of the chart height, rendering within-author variation imperceptible. The per-author independent Y-axis adopted in the final design resolves this tension.

The bar-code view provides three ordering modes (by sector, by frequency, alphabetical) and two scale modes (percentage, absolute mention count), all switchable via controls at the top of the page. A text search field highlights the bars corresponding to a searched place name across all authors simultaneously, and a sector legend allows individual sectors to be hidden, narrowing the display to a subset of the city. Clicking any bar opens a detail panel listing the books that mention the selected place and a representative example sentence, mirroring the sentence-level detail added to the radar chart.

Small multiples (tertiary design). The small multiples view renders five side-by-side maps using Leaflet.js with a CartoDB Positron basemap. Bubble size encodes mention frequency (log scale by default, switchable to linear); bubble colour identifies the author. An optional synchronised-zoom mode locks all five panels to the same viewport, enabling direct spatial comparison. Clicking any bubble opens a popup with the place name, mention count, sector, and an example sentence. A sector-legend filter hides bubbles outside a selected set of sectors across all five panels simultaneously, and a "compare two authors" mode replaces the five-panel grid with two larger, independently zoomable maps selected via dropdown menus, for closer side-by-side inspection of a chosen pair.

The CartoDB Positron style was selected over standard OpenStreetMap tiles because its reduced visual complexity (lighter colours, fewer labels) allows the data bubbles to remain visually prominent. The coordinate range is fixed to the Edinburgh core area as described in Finding 5 above.

3.5 Direction 2: Narrative Topology Networks

Three visualisation designs were developed and implemented for Direction 2.

Force-directed network (primary design). The force-directed network renders the document-level co-occurrence structure as a node-link diagram. Each node represents a place name; node size encodes total mention frequency across all authors. Each edge represents a co-occurrence pair; edge width encodes co-occurrence weight using the function $w' = (w/w_{\max})^2 \times 12$. The squared normalisation was chosen after testing linear and square-root mappings: linear mapping produced edges that were visually indistinguishable in width; square-root mapping improved discrimination but still left the strongest edges insufficiently salient. The squared function produces a clear visual hierarchy while still rendering weaker connections as visible, thin lines. This function was reached after testing two earlier mappings. A linear mapping ($w' = w \times 0.3$) produced edges that were visually indistinguishable across the full weight range 1–28. A square-root mapping ($w' = \sqrt{w/w_{\max}} \times 8$) improved discrimination marginally but left the strongest connections insufficiently prominent relative to weaker ones.

Node positions are determined entirely by the force simulation (Fruchterman-Reingold via D3's `forceSimulation`), with no geographic component. A node's position in the layout reflects its pattern of co-occurrence connections, not its map location. This is a deliberate design choice: the diagram is meant to show narrative topology, and importing geographic position would conflate the two.

An interactive weight-threshold slider (range 1–28) allows the user to progressively reveal weaker connections, from the sparsest high-confidence view (threshold 28, showing only the single strongest pair) to the densest view (threshold 1, showing all 403 pairs). Nodes are draggable, and the diagram supports scroll-to-zoom and pan. Hovering over a node reveals a tooltip listing the five places most strongly co-occurring with it; clicking a node opens a detail panel with the same co-occurrence ranking, the books that mention the place, and an example sentence. A toggle switches node size between the absolute-frequency encoding described above and a percentage-of-total encoding, trading a compressed logarithmic scale for a more literal but more visually dominated-by-outliers area encoding.

Linear connection diagram (secondary design). The linear connection diagram arranges place names along a horizontal axis, ordered left to right by total mention frequency. Bézier arcs connect co-occurring pairs above the axis; arc height is proportional to the horizontal distance between the two connected nodes, and stroke width is proportional to co-occurrence weight. A weight-threshold slider (range 1–28) controls which pairs are shown, and the horizontal ordering can be switched between frequency, sector, and alphabetical. Clicking an arc opens a detail panel listing the books in which both connected places co-occur; hovering over a place reveals an example sentence. A search field highlights a named place across the diagram.

The choice of a linear rather than circular layout followed a specific piece of supervisor feedback: a circular arrangement (as in a chord diagram) implies that the structure being represented is cyclical or symmetrical, which is inappropriate for literary narratives, which have a definite beginning and end. The linear axis does not make this implication. The leftmost position (highest-frequency place) corresponds loosely to centrality in the corpus, and the arrangement from left to right can be read as a rough ordering by narrative salience.

3.5.0.0.1 Designs considered but not selected. Two Direction-1 candidates were explored but ultimately deprioritised. A colour map silhouette design used a simplified Edinburgh coastline with area shading proportional to mention frequency; initial data analysis showed that all five authors concentrate their mentions in the city centre, producing visually similar heatmaps with limited discriminability between authors. A chord diagram was explored for Direction 2 but replaced by the linear connection diagram following supervisor feedback that a circular arrangement implies cyclical or symmetrical structure, inappropriate for literary narratives with a definite beginning and end.

Two further directions considered during early design were prototyped but deferred. A *Reader-plot timeline* would represent the sequence of place-name appearances across a single text on a horizontal narrative axis, directly exploiting the `position_pct` field in the `mention_order` table; the data infrastructure for this direction is complete, and it is identified as future work in Chapter 7. A *literary silences map* would visualise Edinburgh places that never appear in the corpus, following Moretti's concept of literary silences [5]; this direction was deferred on methodological grounds, as the LitLong corpus's coverage is not comprehensive enough to treat absence as meaningful evidence.

Metro-style map (illustrative design). The metro-style map arranges place names as stations on a small number of named lines, following the visual language of the London Underground map. The London Tube map, designed by Harry Beck in 1933, replaced geographic accuracy with functional topology: it told travellers which line to take and where to change, rather than where stations were physically located. This project applies the same substitution to literary geography: the lines represent narrative corridors (clusters of places that co-occur strongly in the same works), not physical routes. Beck's map, however, still encoded a real network—the lines corresponded to the physical tracks trains actually ran on; only the geometry was distorted for legibility. An initial prototype of the literary metro map did not have this property: eight lines were defined manually along geographic and cultural axes (a northern coastal line, an Old Town line along the Royal Mile, and so on), and station order within each line followed geographic sequence rather than measured narrative connection.

An audit of this prototype against the document-level co-occurrence data (Finding 3, Section 3.2) found that only 15 of 67 adjacent same-line station pairs (22%) shared at least one book in common: the map's line structure was a geographic argument wearing the visual grammar of a narrative one. The design was rebuilt around the following pipeline to close this gap.

1. **Community detection.** Modularity-based community detection (`networkx.community.greed` is applied to the 57-place, 403-pair co-occurrence graph (edges with weight ≥ 3), so that line membership is determined by measured narrative connectivity rather than manual geographic grouping. Communities larger than 18 stations are recursively re-clustered on their own induced subgraph, so that one line does not absorb several unrelated sub-clusters. Places with no strong connection to any cluster are folded into whichever cluster they connect to most strongly in the unthresholded graph.

2. **Station ordering.** Within each line, stations are ordered by a nearest-neighbour chain over real edge weights, starting from the strongest internal pair and appending each subsequent station to whichever end of the growing chain it connects to most strongly. This guarantees, by construction, that stations adjacent on a line are backed by measured co-occurrence wherever the underlying graph permits it.

3. **Interchanges.** A pair of stations on different lines with co-occurrence weight ≥ 10 produces a genuine interchange: the station is added as a real stop on the second line, rather than being connected by an illustrative cross-reference. This threshold was chosen empirically to yield a legible number of interchanges (12 of 57 stations); a lower threshold (weight ≥ 6) produced 28 interchanges, judged too dense to read as a small set of meaningful hubs.

4. **Line naming.** Names are assigned automatically: if a single author accounts for at least 55% of the book-mention volume attributed to a line's home stations, the line is named after that author (e.g. "Welsh's Edinburgh"); otherwise it is named after its two highest-mention stations (e.g. "Leith & Forth Corridor").

5. **Layout.** A single shared coordinate is computed for every station using a co-occurrence-weighted spring embedding (`networkx.spring_layout`), so that two lines sharing an interchange meet at one physical point rather than being drawn in separate lanes and cross-referenced. Continuous positions are snapped to an integer grid with a collision-avoidance pass; the one station with zero measured co-occurrence is anchored to its assigned line's centroid rather than left to the spring layout's arbitrary placement of unconnected nodes.

6. **Octilinear correction and label placement.** Each line segment between two grid points is checked against Beck's angle convention (0°/45°/90°); segments that do not already satisfy it are redrawn as a two-part "dog-leg"—a diagonal run followed by a straight run—rather than an arbitrary-angle diagonal. Station labels are placed by a greedy search over eight compass positions at two candidate distances from each station, choosing the first position whose estimated text bounding box does not overlap an already-placed label or station marker.

This pipeline reduces the map to five data-derived lines (rather than eight hand-drawn ones) and raises the proportion of adjacency backed by real co-occurrence from 22% to 88% (46 of 52 adjacent station pairs; per-line figures are reported in

Appendix B.5). The community structure also surfaced two findings not visible in the hand-drawn version, discussed further in Section 6: the word "Leith" clusters with the historical Old Town corridor associated with Scott and Lockhart rather than with the Welsh-dominated periphery line, and a small five-station line—Castle Street, Lasswade, the University of Edinburgh, the Royal Society of Edinburgh, and St Giles—emerged as a coherent biographical corridor corresponding to the places Lockhart's *Life of Walter Scott* associates with Scott's own residences and professional life, rather than with Scott's fiction.

3.6 Design Summary

Table 3.1 summarises the six visualisation designs and their key properties.

Table 3.1: Summary of the six visualisation prototypes.

Direction	Design	Primary encoding	Key interaction
1	Radar chart	Polygon shape across 14 sector axes	Click vertex for sector detail panel; example sentence on hover
1	Bar-code	Bar height = mention share per place	Sector/frequency/alpha ordering; per-author Y-axis; search
1	Small multiples	Bubble size on real map	Sync zoom; log/linear toggle; sector filter; 2-author compare mode
2	Force-directed network	Node size = frequency; edge width = co-occurrence	Weight slider; drag nodes; click node for detail panel; absolute/percentage scale
2	Linear connection	Arc height = node distance; arc width = weight	Weight slider; sort by frequency/sector/alphabet; click arc for book list; search
2	Metro-style map	Line membership = community-detected narrative corridor	Click line to isolate; click station for detail panel; search

Chapter 4

Implementation

4.1 Data Pipeline

Database setup

PostgreSQL 16 was installed via Homebrew on macOS and a database named `litlong_edinburgh` was created. The full LitLong dump (`litlong_2022-06-21.sql`, 145 MB, approximately 2.3 million lines) was imported directly using `psql`. This dump is more complete than the four CSV snapshots provided with the project brief: it contains 620 documents (versus 548 in the CSV), 50,248 mention records (versus 48,056), and 27 tables not present in the CSV export, including `api_sentence` (full source text for every mention), `api_genre` (29 literary genre classifications), and `api_posmention` (2,085,834 part-of-speech annotation records).

One table failed to import from the dump: `api_location`, which uses PostGIS geometry types (`geom`, `poly` fields) that require the PostGIS extension. Since PostGIS was not installed, `api_location` was instead imported from the CSV snapshot using Python/pandas, retaining only the fields `id`, `text`, `lat`, `lon`, `gazref`, and `ptype`, and discarding the PostGIS-dependent geometry fields. These geometry fields are not required for any of the current visualisation directions, which use latitude/longitude centroids throughout.

The database name conflicts with user role assumptions in the dump (which was originally owned by user `palimpsest` on a remote server). Import warnings about missing roles are harmless and do not affect the data.

Narrative order extraction

The api_locationmention table contains a field start_word encoding a global word index for each mention (format: w81489). This field was validated to be strictly monotonically increasing within each document by extracting the integer values and checking for violations across a sample of documents. No violations were found. This property means that start_word can be used directly to reconstruct the reading order of place-name mentions within any document, without parsing the source text.

A custom table mention_order was constructed with the following fields: id (foreign key to api_locationmention), document_id, location_id, word_num (integer extracted from start_word), mention_order (rank of this mention within its document, 1 = earliest), and position_pct (normalised position within the document, computed as $(word\_num - \min)/(\max - \min)$ per document). The position_pct field enables Reader-plot style timeline visualisations and will support future work on narrative sequence analysis.

Sector classification

The sector classification procedure is described in Section 3.3. The implementation uses the shapely Python library (version 2.0.1) for point-in-polygon intersection. The two GeoJSON boundary files were downloaded from the City of Edinburgh Council's ArcGIS REST API:

- Neighbourhood Partnership Areas: Layer ID 28 at edinburghcouncilmaps.info
- Natural Neighbourhoods: Layer ID 188 at the same endpoint

Each place in api_location with a valid latitude and longitude was tested against the 14 sector polygons in order. If a point fell within a polygon, it was assigned to that sector (exact match). If it fell within the Edinburgh bounding box but outside all polygons—due to gaps between polygon boundaries—it was assigned to the nearest polygon by Euclidean distance in projected coordinates (nearest-neighbour assignment). Points outside the bounding box were assigned to a catch-all category, "Outer Scotland," and excluded from sector-based analyses. The results were written to a derived table location_sectors and exported as a CSV for use in the D3 visualisations.

Co-occurrence computation

Document-level co-occurrence between place names was computed in Python using the `itertools.combinations` function. For each document, the set of distinct place names mentioned in that document was extracted; for every pair in that set, the pair's co-occurrence count was incremented by one. The resulting co-occurrence matrix was filtered to retain only pairs with weight ≥ 2 (appearing together in at least two documents), yielding 403 pairs. The top co-occurring pairs were exported as `network_edges.csv` and `network_nodes.csv`.

Sentence and book enrichment

A second pass over the same five-author mention data joins `api_sentence` and `api_document` to attach, for every place, a small sample of full source sentences (deduplicated, truncated to 220 characters, with the contributing book and author) and a ranked list of books that mention it. For Direction 2, the same join is additionally computed per co-occurrence edge, so that the pair of places behind an arc or a network edge can be traced back to the specific books in which both appear. This enrichment step does not alter any previously computed percentage, count, or co-occurrence weight; it is a separate lookup table, keyed by place name (and by place pair for edges), embedded alongside the existing data in each HTML file. Sampling from `api_sentence` surfaces the source text verbatim, including informal register and strong language where present in the original (for example, in Irvine Welsh's Scots-dialect prose); this is an authentic property of the corpus rather than an artefact of the sampling, but it is worth flagging before using these visualisations in a supervisor demonstration or a public-facing study session, since a hover tooltip is not the reader's first encounter with the surrounding literary context.

4.2 Direction 1: Author Spatial Fingerprints — Implementation

Radar chart

The radar chart is implemented using D3's `lineRadial` function with `curveLinearClosed`, which produces a closed polygon across the 14 angular positions. The angular spacing is $2\pi/14$ radians per sector. The radial scale maps sector proportion to a radius between

0 and $R = 250$ pixels, where the scale maximum is set to the observed data maximum (0.436) rather than to 1.0.

Hover interaction is implemented using invisible `circle` elements of radius 14 pixels centred on each vertex. This radius was chosen after testing smaller values: a radius of 6 pixels requires pixel-precise cursor placement and is difficult to target reliably. At radius 14, hovering is comfortable without the targets overlapping each other. On hover, a tooltip shows the sector name and percentage; on click, a detail panel opens showing the top eight place names in that sector for the selected author, ordered by mention count, together with a bar chart and the top five books.

Data is embedded directly as JavaScript arrays in the HTML file. This is necessary because browsers block `XMLHttpRequest` calls to local files under the `file://` protocol; a `d3.csv()` call would silently fail. The same embedding approach is used for all six D3 visualisations.

Bar-code fingerprint

Each author's row is rendered as an SVG element appended to a container div. The Y-axis domain is `[0, maxVal]`, where `maxVal` is the maximum proportion value for that author across all place names—not the global maximum across all authors. This per-author scaling is a deliberate design decision, as justified in Section 3.

Sector boundaries are visualised using two complementary encodings: a low-opacity background rectangle in the sector's colour spans the bars belonging to each sector, and a narrow colour strip below the bar area identifies the sector. Text labels are rendered below the colour strip only when the sector is wide enough to accommodate them without overlap (threshold: 40 pixels); narrower sectors are identified by colour alone. This approach avoids the text-stacking problem that arose in earlier versions when 14 sector labels were displayed regardless of available width.

Switching between ordering modes (sector, frequency, alphabetical) and scale modes (percentage, absolute) is implemented by re-rendering the SVG elements in JavaScript with the new ordering applied to the place-name array, rather than by animating existing elements. This approach is simpler to implement correctly and avoids the complexity of D3 transition management across multiple SVG groups.

Small multiples

Each of the five panels is implemented as an independent Leaflet.js map instance, initialised with the CartoDB Positron tile layer. Place-name bubbles are rendered as Leaflet `CircleMarker` objects, whose `radius` property is recomputed whenever the scale mode changes. The two-stage Leaflet/D3 overlay approach used in earlier prototypes (in which D3 SVG elements were positioned over the map using screen-coordinate projection) was replaced with native Leaflet `CircleMarker` objects, which correctly handle map pan and zoom without requiring manual coordinate projection.

Synchronised zoom is implemented by attaching a `moveend` event listener to each map; when the listener fires and the sync toggle is enabled, it calls `setView` on all other maps with the current centre and zoom level. A boolean `isSyncing` flag prevents the listener from triggering cascading updates.

4.3 Direction 2: Narrative Topology — Implementation

Force-directed network

The simulation uses four forces: `forceLink` (spring attraction between connected nodes, with link distance inversely proportional to co-occurrence weight), `forceManyBody` (uniform repulsion, strength -400), `forceCollide` (collision avoidance with radius equal to node radius plus 18 pixels), and `forceCenter` (gentle attraction to the canvas centre). The simulation is re-run from scratch whenever the weight threshold is changed, because the set of active nodes and edges changes and a smooth transition between graph topologies is not meaningful.

Node labels are positioned at $y = node.y + r + 12$ pixels, placing them below each node. An earlier version positioned labels at the node centroid with white fill, expecting the text to appear inside the node; in practice, the labels were invisible because the white text was rendered against the dark node background. Repositioning labels below the node boundary resolved this issue.

Linear connection diagram

Each arc is drawn as a quadratic Bézier path: `M x1 y0 Q midX (y0 - h) x2 y0`, where `midX` $= (x1 + x2)/2$ and `h` $= |x2 - x1| \times 0.42$. The coefficient 0.42 was chosen to produce arcs that are visually distinct without extending beyond the SVG canvas for the longest connections. Stroke width is computed as $(w/w_{\max})^{1.5} \times 6$.

Metro-style map

The redesigned pipeline described in Section 3.5.0.0.1 is implemented as two Python scripts. `build_metro_lines.py` performs the community detection, station sequencing, interchange detection, and layout, and writes a single JSON file containing, per line, its name, colour, member stations, each station's shared (x,y) grid coordinate, and the adjacency-backed diagnostic count; `build_metro_html.py` consumes this JSON and renders the final HTML file, embedding the data as a JavaScript object literal in the same manner as the other five visualisations. Separating data derivation from rendering allows the underlying clustering parameters (community-detection threshold, interchange threshold, maximum line size) to be adjusted and the map regenerated without touching the rendering code.

Community detection is order-sensitive in an unwanted way: Python's hash-randomised iteration over sets and dictionaries means that `greedy_modularity_communities` can break ties differently between runs, changing the order in which stations are sequenced along a line (though not which line a station is assigned to). Running the script with the environment variable `PYTHONHASHSEED=0` fixes this and was adopted as standard practice for regenerating the dataset.

At render time, each line is drawn as a single SVG path through its stations' shared grid coordinates, converted to pixel coordinates and corrected for octilinearity (Section 3.5.0.0.1) by inserting an intermediate "dog-leg" point wherever two consecutive stations are not already aligned at a multiple of 45°. Because coordinates are shared across lines, two lines that both stop at the same interchange are drawn passing through the same physical point, producing genuine visual crossings rather than the separate parallel lanes used in the initial prototype. Interchange stations (those appearing on two or more lines) are marked with an outer white circle of radius 9 and an inner dark circle of radius 4, following the London Underground convention; all other stations use a smaller white circle with a coloured stroke matching their line's colour. Station labels are positioned by the greedy eight-direction search described in Section 3.5.0.0.1; an initial fixed below-circle placement produced ten overlapping label pairs among the 57 stations in the densest cluster, reduced to two after adopting the greedy search. D3 zoom behaviour is attached to the SVG to support scroll-to-zoom and drag-to-pan; clicking a line or a legend entry isolates it by dimming all other lines and stations, and a text field highlights matching station names.

4.4 Technical Decisions and Constraints

Data embedding. All six D3 HTML files embed their data as JavaScript object literals. The trade-off is larger file sizes and the need to regenerate the HTML file when the underlying data changes. For a research prototype evaluated with a fixed dataset, this is an acceptable constraint.

Versioning. Each HTML file is versioned by maintaining previous versions under filenames such as `radar_v1.html`, `network_v1.html`, etc. This preserves the design evolution as a record for the dissertation and for the supervisor.

Browser compatibility. All files have been tested in Safari and Chrome on macOS. D3.js v7 and Leaflet.js 1.9.4 are both loaded from CDN at runtime, requiring an internet connection.

PostGIS. The PostGIS extension was not installed, and the geometry fields of `api_location` (`geom`, `poly`) are excluded from all analyses. Latitude/longitude centroid coordinates are used throughout. For the current visualisation directions, centroid coordinates are sufficient; more precise polygon representations of place areas would be needed only for overlay analyses that are not part of the current scope.

4.5 Combined Interface

Chapter 5

Evaluation

5.1 Study Design

The user study is designed to evaluate whether the six visualisation prototypes support the tasks implied by the two research questions: author identification (RQ1) and narrative-relationship identification (RQ2).

Participant groups. Two groups are recruited: domain experts (literary scholars, digital humanists, and researchers with familiarity with Edinburgh's literary geography) and general public (participants without specialist knowledge of Edinburgh literature or information visualisation). Recruiting two groups allows the study to test the hypothesis that expert and non-expert users differ in which design they find most legible, and in whether they spontaneously perceive the divergence between narrative and geographic proximity.

Format. The study is conducted online to maximise recruitment reach and sample size. An online format also ensures that participants interact with the visualisations in a browser environment that reflects real-world usage conditions.

Introduction protocol. Participants are shown each visualisation with only a brief description of the project's purpose ("We are studying how literary place names can be visualised") but without an explanation of what each visual encoding means. This "cold" presentation is methodologically important: it tests whether the designs are self-explanatory, rather than testing participants' ability to apply an explanation they have been given. Participants who require the designs to be explained before they can use them are, in effect, revealing a usability problem.

Tasks. Two task types are used. Task 1 (fingerprint direction): participants are shown a fingerprint visualisation (radar chart, bar-code, or small multiples) and asked which of the five authors they believe it represents. This is a forced-choice identification task; accuracy is the primary metric. Task 2 (topology direction): participants are shown a topology visualisation (force-directed network, linear connection diagram, or metro map) and asked to identify the two places that appear most strongly connected, and to judge whether those two places are geographically close or far apart in Edinburgh. The secondary question tests whether the visualisation spontaneously communicates the narrative/geographic divergence finding. After completing all tasks, participants vote for the most intuitive design in each direction.

Ethics. The study has received ethics approval from the Informatics Research Ethics Committee (application number 446635). Participants complete an online consent form before beginning the study. The participant information sheet and consent form are included in Appendix A.

5.2 Participants

5.3 Study Procedure

5.4 Results

5.5 Discussion of Results

Chapter 6

Discussion

6.1 Design Improvements after User Study

6.2 Narrative Topology vs Geographic Topology

The most striking finding from the data analysis is the co-occurrence of Leith and Princes Street across 28 distinct literary works—the strongest co-occurrence pair in the entire corpus. Geographically, these two places are separated by approximately 2.5 kilometres. Leith is a port district historically associated with trade, working-class communities, and political radicalism; Princes Street is the principal commercial artery of the Georgian New Town, associated with bourgeois Edinburgh. On a geographic map, they appear as two points in the same city. In the narrative topology visualisation, they are the closest pair in the entire network.

This is not coincidental. The journey between Leith and Princes Street is a recurrent structuring device in Edinburgh fiction precisely because it traverses a social as well as a spatial boundary. Irvine Welsh's *Trainspotting* makes this explicit: the characters move between Leith and the city centre as a movement between social worlds, between the margins and the mainstream of Edinburgh life. Walter Scott's historical novels stage similar journeys between the Old Town closes and the wider city as crossings between historical strata. The narrative topology visualisation makes this pattern visible in a way that a geographic map cannot: it shows that these two places are *narratively adjacent*, regardless of their spatial distance.

This finding speaks directly to the theoretical framing of the project. Westphal's geocriticism argues that literary space is relational and constructed, not a reflection of physical geography [11]. The Leith/Princes Street co-occurrence is a concrete, quantitative instance of this argument: the places that are most strongly connected in Edinburgh's literary imagination are not the places that are nearest to each other on a map.

The metro-map redesign (Section 3.5.0.0.1) provides a second, independent test of this claim, and a methodological one: if narrative topology genuinely diverges from geography, a line-grouping derived purely from co-occurrence weights should look different from one derived by hand along geographic axes, and it should be checkable against the data in a way a hand-drawn grouping is not. The audit bore this out—only 22% of the hand-drawn map's same-line adjacencies were backed by real co-occurrence, against 88% once lines were derived from the co-occurrence graph itself—which is itself evidence that geographic intuition is an unreliable guide to narrative structure, consistent with Finding 4 rather than a separate result. The redesign also surfaced two findings that the geographic version could not have produced, because geographic grouping cannot detect them. Leith, considered as a place name across all five authors combined, groups with the historical Old Town corridor associated with Scott and Lockhart rather than with the contemporary, working-class Leith of Welsh's novels: at the level of the aggregated five-author corpus, the name's narrative associations are pulled more strongly toward nineteenth-century historical writing than toward the corpus's single most Leith-saturated author. This is a finding about how a place name functions across a corpus, not about any one author's usage of it, and it is only visible once co-occurrence, rather than an author's own usage, is used to group places. Separately, a five-station corridor—Castle Street, Lasswade, the University of Edinburgh, the Royal Society of Edinburgh, and St Giles—emerged as a coherent cluster corresponding to Lockhart's biographical rather than fictional geography of Scott: these are places Scott lived and worked, as documented in *Life of Walter Scott*, not places his novels are set. The community-detection step distinguishes this biographical register from the historical-fiction register of the neighbouring Old Town corridor without being told to.

6.3 Author Spatial Languages

The near-total absence of shared place names across the five test authors—zero places shared by three or more, only 18 shared by any two—is a finding that can be stated as a simple number but deserves fuller interpretation. It means that each author has constructed, across their body of work, an Edinburgh that is almost entirely their own. McCall Smith's Edinburgh is bounded by the triangle formed by Scotland Street, Dundas Street, and Drummond Place in the New Town. Welsh's Edinburgh is bounded by Leith. Scott's Edinburgh is centred on the Canongate and the Old Town closes. Stevenson's Edinburgh extends south towards the Pentlands, reflecting his childhood at Swanston Cottage, and west towards the Firth of Forth, reflecting his father's lighthouse engineering work.

The radar chart and bar-code fingerprint make these private geographies comparable at a glance. A user who knows nothing of these authors' biographies can see, from the shape of the radar chart polygon, that McCall Smith and Welsh occupy different spatial territories within the same city. The visualisation encodes, in its geometry, a literary-geographic argument that would otherwise require paragraphs of prose.

This near-total vocabulary separation also explains a non-obvious pattern observed when the network graph was scale-tested against the full corpus (`network_scale_explore.html`). Co-occurrence density does not fall smoothly as the number of authors grows: it is high both at the very small end (two authors, or a document set drawn from two or three books) and low at the very large end (all 408 authors), for two different reasons. At the small end, document-level co-occurrence approaches its degenerate case—almost any two places mentioned anywhere in the same handful of books will co-occur at least once, because a single book's own place vocabulary trivially satisfies "appears in the same document." Restricting the network to three books (two by McCall Smith, one by Welsh) produced 152 places and 5,889 co-occurring pairs at weight ≥ 1 (51% of all possible pairs); raising the threshold to weight ≥ 2 collapsed this to 33 places and 424 pairs, almost all of them connecting McCall Smith's two books to each other rather than to Welsh's, because his fictional Scotland Street recurs across his own works while Welsh's Leith does not recur in either of McCall Smith's—a direct, book-level instance of the vocabulary separation just described. At the large end, adding authors adds nodes far faster than it adds genuine cross-author edges, for the same underlying reason, so the 408-author graph is large (533 places, 23,380 pairs) but comparatively sparse (16.5% of possible pairs, against 79.6% for two authors). The five-author baseline used throughout this dissertation sits between these two failure modes: enough authors for a cross-author edge to be a meaningful, comparatively rare signal, rather than either the near-certainty of the two-author or two-book case or the near-absence of the full-corpus case.

6.4 Limitations and Open Questions

Corpus size confound. McCall Smith contributed 17 works to the corpus, Stevenson 16, Scott 12. Welsh contributed only 8, Lockhart 6. Because sector distributions are normalised as proportions of each author's total mentions, a large corpus and a small corpus with identical spatial habits would produce identical fingerprints. However, this normalisation also means that the results are best interpreted as measures of *spatial attention*—what fraction of each author's narrative geography is allocated to each sector—rather than absolute output. The consistency of the spatial patterns across works within each author's corpus (which is visible when individual books are examined) suggests that these are genuine authorial preferences rather than corpus-size artefacts.

Five-author prototype, tested at scale. The current system uses five test authors, selected for contrast and representativeness. Whether this scales to the full corpus was tested directly rather than left as an open question, for all six designs, using a set of scale-exploration tools kept separate from the six canonical five-author prototypes (radar, network, and linear are live interactive pages that recompute their data client-side for any of the 408 authors with location data; bar-code is likewise live; small multiples and metro required batch or capped-live testing for reasons specific to each design, detailed below). The headline result is that the six designs do not share one scaling bottleneck: each fails in a different, specific way, summarised in Table B.6 (Appendix B.6).

Radar and bar-code (Direction 1) both degrade by losing the "compare everything at a glance" property, but through different mechanics. Radar's static polygon overlay becomes visually unreadable well before 408 authors (individual shapes are no longer distinguishable by 20), though a hover-to-isolate interaction recovers single-author legibility at any N at the cost of losing the all-at-once view. Bar-code never becomes unreadable or slow—408 rows render in 20 milliseconds—but the page becomes roughly 50 screens tall, so comparing two arbitrary authors requires scrolling between points far apart rather than looking at two adjacent shapes.

Network and linear (Direction 2) share a root cause—one mark per co-occurring place pair, laid out along a single spatial dimension with no wrapping—but manifest it on different axes: network's force-directed layout degrades gracefully (only the graph's dense core becomes unreadable; the periphery stays legible even at 408 authors), while linear's fixed horizontal axis simply grows the canvas to tens of thousands of pixels wide, so that a screenshot of the visible (scrolled) viewport looks almost unchanged from the five-author version even though the underlying canvas is roughly 27 screen-widths across.

Small multiples is the only design with a measured, genuine computational cost rather than a purely visual or navigational one, because each panel is a live map instance requesting real tiles: initialisation time per panel is roughly constant from 5 to 50 authors ($\approx$1–3ms each) but degrades by a factor of 3.6 from 50 to 408 (measured directly, not extrapolated, at 1.47 seconds total), a real main-thread delay a user would feel as a freeze, distinct from the other designs' purely visual degradation.

Metro is the only design where the failure is in the underlying algorithm rather than the picture: re-running the community-detection pipeline (Section 3.5.0.0.1) at larger N shows the adjacency-backed percentage declining monotonically (96% at 2 authors, 92% at 5, 84% at 20, 72% at 50) while, at 50 authors, one community grows to 75 stations because `greedy_modularity_communities` judges it internally cohesive enough that the recursive size cap cannot split it further—a structural limit of the clustering approach, not a rendering or layout problem.

Metro-map layout is heuristic, not optimal. The metro-map redesign (Section 3.5.0.0.1) replaces manual line assignment with community detection and replaces manual station placement with a spring embedding corrected for octilinearity, but neither step solves the underlying schematic-layout problem exactly—Nöllenburg and Wolff's result [6] establishes that an exact solution is computationally hard in general. Two residual limitations follow directly from this. First, the line structure is sensitive to the community-detection threshold and the five-author, top-15-places-per-author filter used to build the co-occurrence graph; this was confirmed directly rather than left as speculation (Appendix B.6): re-running the pipeline at 20 and 50 authors produces 8 and 11 lines respectively, with the proportion of adjacency-backed adjacent stations falling from 92% at 5 authors to 72% at 50, and at 50 authors one community grows to 75 stations because `greedy_modularity_communities` treats it as internally unsplittable despite exceeding the 18-station recursion cap—a structural limit of the clustering approach that the current five-line map does not exhibit only because it sits at a favourable point on this curve, not because the underlying method guarantees it. The current five lines should therefore not be read as a fixed taxonomy. Second, the greedy label-placement search leaves two of 57 station labels overlapping in the single densest interchange cluster, which a fully general solver (or manual adjustment, as is common practice even for professionally drawn transit maps) could resolve.

PostGIS and polygon precision. The `api_location` table records places as latitude/longitude centroids, not polygons. For a place like "Old Town," the centroid may not accurately represent the full extent of the area referenced in the text. PostGIS polygon data is available in the original database schema but requires the PostGIS extension, which was not installed in this project. Future work could install PostGIS and use the polygon representations for more precise sector assignment.

Source text sampling. All six prototypes now surface example sentences drawn from `api_sentence` (Section 4), addressing the supervisor's priority that readers should be able to see the literary sentence in which a place appears rather than only an aggregate count. The remaining limitation is that examples are sampled (with a fixed random seed, for reproducibility) rather than curated: a place with many mentions is shown a small, arbitrary subset of its sentences, which may not be the most illustrative or representative ones. A curated or relevance-ranked selection is left for the combined interface.

Reader-sketch and narrative time. The supervisor noted a direction labelled "Reader-sketch?" in early meetings, referring to visualisations that represent the reading experience—the sequence in which places are encountered as the reader moves through a text. The `position_pct` field in `mention_order` provides the necessary data for this, and Reader-plot timeline visualisations were prototyped but not included in the user study. This remains an open direction.

Chapter 7

Conclusion

This dissertation has argued that place names in literary text are data about narrative structure, not merely about geographic location, and has demonstrated that this argument can be operationalised as a set of interactive visualisations grounded in real corpus data. The LitLong Edinburgh database—620 literary works spanning three centuries, 2,135 place names, 50,248 mention records—provided the empirical foundation. Six visualisation prototypes were developed across two research directions, implemented in D3.js, and evaluated through a user study.

The project's contributions include: a validated spatial classification of Edinburgh place names using official City of Edinburgh Council administrative boundary data; an empirical finding that each of five representative Edinburgh authors constructs an almost entirely private imaginary geography of the city; a method for extracting narrative reading order from the LitLong database using the `start_word` field as a monotonic index; and a suite of interactive visualisation prototypes that make authorial spatial patterns and narrative co-occurrence structure visible in forms not previously available.

Several directions for future work are identified. Scaling the system to the full 424-author corpus would require a combined interface with author-selection controls and a layout strategy for the radar chart and network visualisations that remains legible at large author counts. Integrating the `api_sentence` source text into the interactive detail panels would provide the most direct form of "details on demand" and would ground the abstract visualisation encodings in the literary language itself. The `api_posmention` table, containing over two million part-of-speech annotations, would support a third research direction: narrative weight analysis, examining whether a place name appears in dialogue, narration, or description, and whether this varies systematically across authors or genres. Finally, the concept of "literary silences"—Edinburgh places that appear in the gazetteer but never in any literary work—would complement the current analysis by revealing what Edinburgh's literary imagination systematically excludes.

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Appendix A

Study Material

A.1 Participant Information Sheet

A.2 Consent Form

A.3 Task Questions

A.4 Survey Questions

Appendix B

Data and Implementation

B.1 Database Schema

The LitLong Edinburgh PostgreSQL database contains 27 tables after full import. The tables most relevant to this project are listed in Table B.1.

Table B.1: Key tables in the LitLong Edinburgh database.

Table	Rows	Description
`api_document`	620	Literary works (title, publication date, type)
`api_author`	424	Authors (name, gender, Open Library ID)
`api_document_author`	1,232	Many-to-many: documents ↔ authors
`api_location`	2,135	Place names (text, latitude, longitude)
`api_locationmention`	50,248	Mention records (start_word, page_id, sentence_id)
`api_sentence`	50,248	Full source text for each mention
`api_genre`	29	Literary genre classifications
`api_posmention`	2,085,834	Part-of-speech annotations
`mention_order`	50,248	Custom: narrative order and position (derived)

B.2 Sector Definitions

Table B.2 lists the 14 sectors used in the spatial classification, their data sources, and their place-name counts.

Note: Nearest-neighbour counts are included in the exact-match totals above. Source: City of Edinburgh Council Open Spatial Data Portal (Open Government Licence v3.0). "Copyright

Table B.2: The 14 spatial sectors and their place-name counts.

Sector	Source	Exact matches	Nearest-neighbour
Old Town	Natural Neighbourhoods	296	—
South Central	NP Areas	255	—
New Town	Natural Neighbourhoods	217	—
Leith	NP Areas	152	—
Inverleith	NP Areas	140	—
Craigentinny/Duddingston	NP Areas	73	—
Pentlands	NP Areas	66	—
Canongate	Natural Neighbourhoods	64	—
Almond	NP Areas	64	—
South West	NP Areas	44	—
Western Edinburgh	NP Areas	43	—
Portobello/Craigmillar	NP Areas	38	—
Liberton/Gilmerton	NP Areas	36	—
Forth	NP Areas	27	—
Outer Scotland (excluded)	—	232	—

City of Edinburgh Council, contains Ordnance Survey data © Crown copyright.”

B.3 Author Selection

The five test authors were selected on the basis of mention count, number of works in the corpus, and thematic contrast. Table B.3 summarises their corpus statistics and dominant sectors.

Table B.3: Five test authors and their corpus statistics.

Author	Mentions	Works	Dominant sector	%
Alexander McCall Smith	7,320	17	New Town	43.6
Irvine Welsh	2,634	8	Leith	44.3
John Gibson Lockhart	3,046	6	New Town	25.9
Walter Scott	2,796	12	Old Town	32.0
Robert Louis Stevenson	1,778	16	Old Town	20.7

B.4 Co-occurrence Analysis

Table B.4 lists the ten strongest document-level co-occurrence pairs in the corpus, computed across the five test authors.

Table B.4: Top 10 document-level co-occurrence pairs (five test authors).

Place A	Place B	Weight (documents)
Leith	Princes Street	28
New Town	Princes Street	18
Dundas Street	Princes Street	17
Bruntsfield	Dundas Street	16
Bruntsfield	Princes Street	16
Dundas Street	New Town	16
Dundas Street	Morningside	15
Morningside	Stockbridge	15
Princes Street	Stockbridge	15
Dundas Street	The Meadows	15

B.5 Metro-Map Line Composition

Table B.5 lists the five lines produced by the community-detection pipeline described in Section 3.5.0.0.1, with the number of home stations (stations whose narrative cluster this line is), the total station count including spliced-in interchange stops, total mentions summed across home stations, and the adjacency-backed diagnostic (the number of home-station-adjacent pairs, out of the total possible, that share at least one book in common).

Table B.5: The five metro-map lines and their composition.

Line	Home stations	Total stops	Mentions	Adjacency backed
Smith's Edinburgh	15	16	2,786	13/14 (93%)
Leith & Forth Corridor	12	23	1,564	9/11 (82%)
Scott's Edinburgh	13	14	826	9/12 (75%)
Welsh's Edinburgh	12	12	494	11/11 (100%)
Lockhart's Edinburgh	5	5	202	4/4 (100%)
Total	57	—	5,872	46/52 (88%)

Note: "Total stops" exceeds "home stations" where a line has been joined by an interchange station whose home cluster is a different line (12 stations are interchanges across the whole map). The 22% adjacency-backed figure for the earlier, hand-drawn eight-line version was computed over 67 same-line adjacent pairs, 15 of which shared a book.

B.6 Scale Exploration Summary

All six designs were re-tested at four benchmark author counts (2, 5, 20, and either 408 or the largest value practical for the design in question), plus a 2–3 book test for network and linear, using scale-exploration tools kept separate from the six canonical five-author prototypes (`data/processed/scale_exploration/`). Table B.6 summarises the failure mode identified for each design; the full quantitative detail (per-N tables, timing measurements, and the decision log explaining every scope choice) is recorded in `data/processed/scale_exploration/NOTES.md` rather than reproduced in full here.

Table B.6: Scaling failure mode identified for each of the six designs.

Design	Failure mode	Key evidence
Radar	Visual overlap	Polygons indistinguishable by N=20; unreadable at 408; hover-to-isolate recovers single-author legibility at any N.
Bar-code	Navigation cost	408 rows render in 20ms (no rendering problem) but the page is ≈45,000px tall—about 50 screens of scrolling.
Network	Visual overlap (graceful)	Core becomes a dense mass at 408 authors; periphery stays legible. Density is U-shaped in N (79.6% at 2 authors, ≈25% at 5, 16.5% at 408), driven by vocabulary non-overlap (Finding 1), not raw node count.
Linear	Horizontal sprawl	Canvas width grows from 1,100px (5 authors) to 29,790px (408 authors); a screenshot of the scrolled viewport looks unchanged because the same high-mention places sort leftmost regardless of N.
Small multiples	Computational cost (the only one)	Live Leaflet map initialisation degrades 3.6× per-panel from 50 to 408 authors (1.47s total at 408, measured directly); the one design where scale risks a real freeze, not just illegibility.
Metro	Clustering algorithm breakdown	Adjacency-backed proportion falls monotonically (96%→72%, N=2 to 50); at N=50 one community reaches 75 stations because modularity detection judges it unsplittable despite exceeding the 18-station cap.